

Probability Final Exam

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1. $(a \times b \times d)$

2. (c)

3. $M_Z(t) = E\left(e^{t(\lambda_1 Y_1 - \lambda_2 Y_2)}\right)$

$\left(\because Y_1, Y_2 \text{ are indep}\right) \Downarrow E\left(e^{(\lambda_1 t) Y_1}\right) E\left(e^{(-\lambda_2 t) Y_2}\right)$

$$= M_{Y_1}(\lambda_1 t) M_{Y_2}(-\lambda_2 t)$$

$$\text{Now } M_{Y_1}(t) = \left(\frac{\lambda_1}{\lambda_1 - t}\right) \mathbb{1}_{\{t < \lambda_1\}}$$

$$M_{Y_2}(t) = \left(\frac{\lambda_2}{\lambda_2 - t}\right) \mathbb{1}_{\{t < \lambda_2\}}$$

$$\therefore M_Z(t) = \left[\left(\frac{\lambda_1}{\lambda_1 - t}\right) \mathbb{1}_{\{\lambda_1 - t < \lambda_1\}} \right] \left[\left(\frac{\lambda_2}{\lambda_2 - (-\lambda_2 t)}\right) \mathbb{1}_{\{-\lambda_2 t < \lambda_2\}} \right]$$

$t < 1$ $t > -1$

$$= \left(\frac{1}{1-t}\right) \left(\frac{1}{1+t}\right) \mathbb{1}_{\{-1 < t < 1\}} = \left(\frac{1}{1-t^2}\right) \mathbb{1}_{\{-1 < t < 1\}}$$

This matches the mgf of the double exponential distribution with $b=1$.

By the uniqueness of mgf, we conclude that's the distribution of Z .

4. By Poisson Process's property,

$X \sim \text{Poisson}(z\lambda)$, which has

Variance $z\lambda$.

5. (a)
$$f_Y(y) = \int_{-\infty}^{\infty} f_{X,Y}(x,y) dx$$

$$= \int_0^2 \frac{1}{6}(x+4y) dx \quad (\text{for } 0 < y < 1)$$

$$= \frac{1}{6} \left[\frac{x^2}{2} + 4xy \right]_{x=0}^{x=2}$$

$$= \begin{cases} \frac{1}{3}(1+4y) & \text{if } 0 < y < 1 \\ 0 & \text{o.w.} \end{cases}$$

$$5 \text{ (b)} \quad f_{X|Y}(x|y) = \begin{cases} \frac{\frac{1}{6}(x+4y)}{\frac{1}{3}(1+4y)} & 0 < x < 2, \\ 0 & \text{o.w} \end{cases} \quad 0 < y < 1$$

$$\therefore f_{X|Y=0.25}(x|0.25) = \frac{\frac{1}{2}(x+1)}{(1+1)} \quad (\text{for } 0 < x < 2)$$

$$= \frac{x+1}{4}$$

$$\therefore P(X \leq 1) = \frac{1}{4} \int_0^1 (x+1) dx = \frac{3}{8}$$

6. (a)

7. Let $A = \{\text{the ball drawn from Box 1 is red}\}$
 $B = \{\text{the ball drawn from Box 2 is red}\}$

$$(a) \quad P(A) = \frac{4}{4+2} = \frac{2}{3}$$

$$7(b) \quad P(B) = P(B|A)P(A) + P(B|A^c)P(A^c)$$

$$= \frac{4}{4+3} \cdot \frac{2}{3} + \frac{3}{3+4} \cdot \frac{1}{3}$$

$$= \frac{11}{21}$$

$$(c) \quad P(A|B) = \frac{P(B|A)P(A)}{\underbrace{P(B|A)P(A) + P(B|A^c)P(A^c)}_{\frac{11}{21} \text{ from (b)}}$$

$$= \frac{\frac{8}{21}}{\frac{11}{21}} = \frac{8}{11}$$

$$(d) \quad P(A \cap B) = P(B|A)P(A) = \frac{8}{21}$$

8. Assume that the first event of Type II arrives at T .

$$T \sim \text{Exp}(\lambda_2) \quad E(T) = \frac{1}{\lambda_2}$$

Let $X = \#$ events of Type I prior to T

$$(X|T) \sim \text{Poisson}(\lambda_1, T)$$

$$E(X) = E(E(X|T)) = E(\lambda_1 T) = \frac{\lambda_1}{\lambda_2}$$

9. (b)

10. (a)

11. (c)

12. $X \sim \text{Binomial}(n, p)$

so $E(X) = np$, $V(X) = np(1-p)$

$$E\left(\frac{X}{n}\right) = \frac{1}{n} E(X) = p$$

$$V\left(\frac{X}{n}\right) = \frac{1}{n^2} V(X) = \frac{p(1-p)}{n}$$

13. (a) By the multivariate normal distribution's property, (X_1, X_2) has the bivariate normal distribution with mean $\begin{pmatrix} 1 \\ -2 \end{pmatrix}$ and covariance matrix $\begin{pmatrix} 1 & -0.3 \\ -0.3 & 1 \end{pmatrix}$. We simply drop any X_3 related parts from the original μ, Σ .

13 (b) Since $\text{cov}(X_1, X_2) = -0.3$

is nonzero, X_1, X_2 are uncorrelated.

which implies they are not independent
given that (X_1, X_2) is bivariate normal.

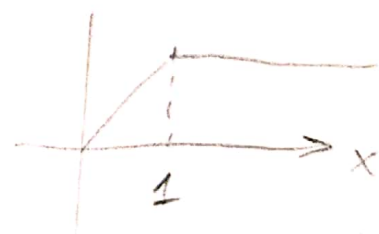
14 (a) (ii)

$$(b) P(H) = \sum_{i=1}^5 P(H|C_i) P(C_i)$$

$$= \frac{1}{5} \sum_{i=1}^5 p_i = \frac{1}{5} \sum_{i=1}^5 \frac{(i-1)}{4} = \frac{1}{2}$$

15 (a)

$$F_X(x) = \begin{cases} 0 & x \leq 0 \\ x & 0 < x < 1 \\ 1 & x \geq 1 \end{cases}$$



$$F_Y(y) = \begin{cases} 0 & y \leq 0 \\ y & 0 < y < 1 \\ 1 & y \geq 1 \end{cases}$$

$$P(Z > z) = P(\min(X, Y) > z)$$

$$= P((X > z) \cap (Y > z))$$

$$\left(\begin{array}{l} \because X, Y \\ \text{are indep} \end{array} \right) \Rightarrow P(X > z) P(Y > z)$$

$$= (1 - F_X(z)) (1 - F_Y(z))$$

$$= \begin{cases} 1 & \text{if } z \leq 0 \\ (1-z)^2 & \text{if } 0 < z < 1 \\ 0 & \text{if } z \geq 1 \end{cases}$$

$$15(b) \quad f_Z(z) = \frac{d}{dz} F_Z(z) = \frac{d}{dz} (1 - P(Z > z))$$

$$= \begin{cases} \frac{d}{dz} (1 - (1-z)^2) & \text{if } 0 < z < 1 \\ 0 & \text{o.w.} \end{cases}$$

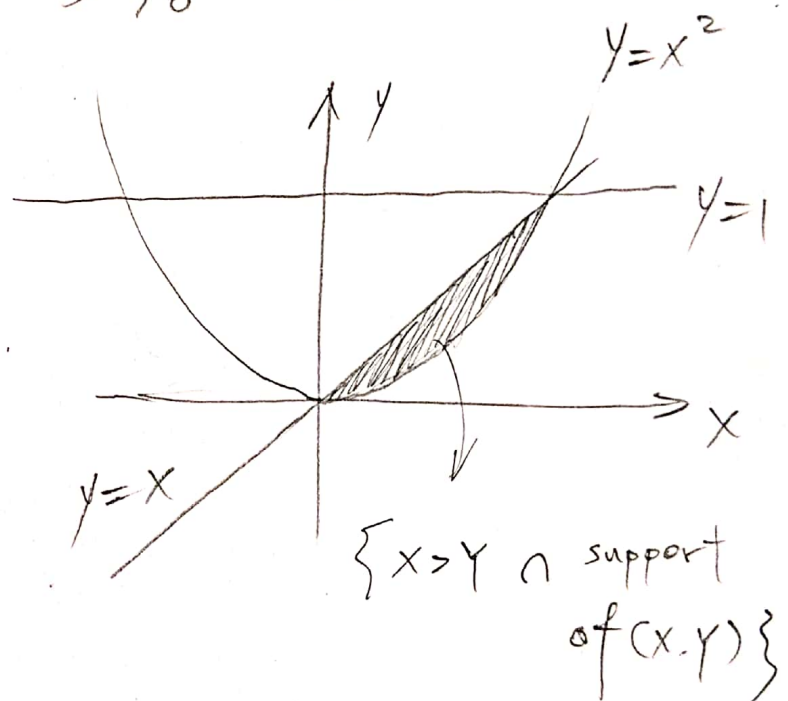
$$= \begin{cases} 2 - 2z & \text{if } 0 < z < 1 \\ 0 & \text{o.w.} \end{cases}$$

$$15(c) \quad E(Z) = \int_{-\infty}^{\infty} z f_Z(z) dz$$

$$= \int_0^1 z(2-2z) dz = \int_0^1 2z - 2z^2 dz$$

$$= 2 \left[\frac{z^2}{2} - \frac{z^3}{3} \right]_0^1 = \frac{1}{3}$$

16.



$$P(X > Y) = \iint_{\{X > Y\}} f_{X,Y}(x,y) dx dy$$

$$= \int_{x=0}^{x=1} \int_{y=x^2}^{y=x} \left(\frac{21}{4}\right) x^2 y dy dx$$